Mock diagnostic examination - Maths full solutions

Full solutions

Use these solutions only after completing the 90-minute attempt. Review each method, result and common mistake, then record the first cause of any lost mark.

Section M0 — short diagnostic questions

Question 1

Part A — `MOCK-M0-Q01-PA` (M1) Method and conditions. Negate the universal quantifier and the inequality, then exhibit one real number for which the negated inequality holds.

The negation is

$$\exists x \in \mathbb{R} \quad \text{such that} \quad x^2 < x.$$

For example, $x = 1/2$ gives $x^2 = 1/4 < 1/2 = x$.

Final result. The displayed existential statement is the negation, and $x = 1/2$ is a counterexample.

Marks (2). Correct existential negation: 1. Valid counterexample with the inequality checked: 1.

Common mistakes. Keeping $\forall$ in the negation; replacing $\geq$ by $\leq$ rather than by $<$; naming a value without checking the inequality.

Question 2

Part A — `MOCK-M0-Q02-PA` (M2) Method and conditions. Establish the logarithms' real domain before combining them.

The two logarithms require $x - 1 > 0$ and $x + 1 > 0$, so $x > 1$. On this domain,

$$\log((x-1)(x+1)) = \log 8 \quad \Longrightarrow \quad x^2 - 1 = 8.$$

Thus $x = \pm 3$, but only $x = 3$ satisfies the domain restriction.

Final result.

$$\boxed{x = 3}.$$

Marks (2). Correct domain and logarithm combination: 1. Correct admissible solution after rejecting $x = -3$: 1.

Common mistakes. Omitting the domain; writing $\log(x-1) + \log(x+1) = \log(2x)$; retaining the extraneous root $x = -3$.

Question 3

Part A — `MOCK-M0-Q03-PA` (M3) Method and conditions. Use the first-order Taylor approximation $f(x) \approx f(a) + f'(a)(x - a)$ about $a = 4$.

Since $f(4) = 2$ and $f'(4) = 1/(2\sqrt{4}) = 1/4$,

$$\sqrt{x} \approx 2 + \frac{1}{4}(x - 4).$$

At $x = 4.4$,

$$\sqrt{4.4} \approx 2 + \frac{1}{4}(0.4) = 2.1.$$

Final result. The approximation is $\sqrt{x} \approx 2 + (x-4)/4$, giving $\boxed{\sqrt{4.4} \approx 2.1}$.

Marks (2). Correct linear approximation about 4: 1. Correct substitution and numerical approximation: 1.

Common mistakes. Centring the approximation at zero; omitting the $(x-4)$ term; using $f'(4) = 1/2$.

Question 4

Part A — `MOCK-M0-Q04-PA` (M6) Method and conditions. A Jacobian has one row for each output and one column for each input. Under the course convention, a gradient is a column vector.

Because $\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^2$, $\mathbf{J}_{\mathbf{F}}(\mathbf{x})$ is 2×3. Because $h : \mathbb{R}^2 \to \mathbb{R}$, $\nabla h(\mathbf{u})$ is 2×1.

Final result.

$$\boxed{\dim \mathbf{J}_{\mathbf{F}} = 2 \times 3, \qquad \dim \nabla h = 2 \times 1}.$$

Marks (2). Correct Jacobian dimension: 1. Correct column-gradient dimension: 1.

Common mistakes. Reversing the Jacobian dimensions; treating the gradient as a row vector; counting the scalar output of h as the gradient length.

Question 5

Part A — `MOCK-M0-Q05-PA` (M8) Method and conditions. Represent the form with its symmetric matrix and apply Sylvester's criterion.

$$q(x, y) = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

The leading principal minors are $2 > 0$ and $4 - 1 = 3 > 0$. Sylvester's criterion therefore shows that the matrix is positive definite.

Final result. The quadratic form is $\boxed{\text{positive definite}}$.

Marks (2). Valid definiteness test or calculation: 1. Correct classification: 1.

Common mistakes. Using 2 rather than 1 in each off-diagonal entry; giving a classification without a supporting test; confusing positive definiteness with positive semidefiniteness.

Section M1 — linear algebra

Question 6 — systems, rank and parameter cases

Part A — `MOCK-M1-Q06-PA` (M4) Method and conditions. Eliminate x from the last two equations and separate the cases $a \neq 1$ and $a = 1$.

Subtracting the first equation from the second and third gives

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & a-1 & 0 & b-1 \\ 0 & 0 & a-1 & b-1 \end{array}\right].$$

If $a \neq 1$, the coefficient matrix has rank 3, so the solution is unique:

$$y = z = \frac{b-1}{a-1}, \qquad x = 1 - y - z = \frac{a+1-2b}{a-1}.$$

If $a = 1$ and $b = 1$, the final two rows vanish. The coefficient and augmented ranks are both 1, so there are multiple solutions:

$$\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 1\}.$$

If $a = 1$ and $b \neq 1$, a reduced row asserts $0 = b - 1$. The coefficient rank is 1 and the augmented-matrix rank is 2, so the system is inconsistent.

Final result.

$$\boxed{\begin{cases} a \neq 1 & : \text{unique solution } \left(\dfrac{a+1-2b}{a-1}, \dfrac{b-1}{a-1}, \dfrac{b-1}{a-1}\right), \\ a = 1,\ b = 1 & : \text{multiple solutions, with both ranks equal to 1,} \\ a = 1,\ b \neq 1 & : \text{no solution, with ranks 1 and 2.} \end{cases}}$$

Marks (8). Correct elimination: 1. Correct unique-case classification: 1. Correct y and z: 1. Correct x: 1. Correct multiple-solution case and solution set: 1. Correct two ranks in that case: 1. Correct no-solution case: 1. Correct two ranks in that case: 1.

Common mistakes. Dividing by $a - 1$ without treating $a = 1$; failing to separate $b = 1$ from $b \neq 1$; confusing coefficient rank with augmented rank; reporting a unique-case formula that does not satisfy the first equation.

Question 7 — eigenvalues, eigenvectors and diagonalisation

Part A — `MOCK-M1-Q07-PA` (M5) Method and conditions. Use triangularity for the eigenvalues, solve each null-space equation, and use the resulting eigenvectors as the columns of $\mathbf{P}$.

The eigenvalues are 4 and 1. For $\lambda = 4$,

$$(\mathbf{A} - 4\mathbf{I}_2)\mathbf{v} = \mathbf{0} \quad \Longrightarrow \quad v_2 = 0,$$

so

$$E_4 = \operatorname{span}\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}.$$

For $\lambda = 1$, $3v_1 + v_2 = 0$, hence

$$E_1 = \operatorname{span}\left\{\begin{bmatrix}1\\-3\end{bmatrix}\right\}.$$

Eigenvectors corresponding to distinct eigenvalues are linearly independent. Equivalently, the determinant of the matrix with these vectors as columns is $-3 \neq 0$, so they form a basis of $\mathbb{R}^2$. One valid diagonalisation is

$$\mathbf{P} = \begin{bmatrix}1 & 1\\0 & -3\end{bmatrix}, \qquad \mathbf{D} = \begin{bmatrix}4 & 0\\0 & 1\end{bmatrix}, \qquad \boxed{\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}}.$$

Final result. The eigenspaces and one consistent pair $(\mathbf{P}, \mathbf{D})$ are displayed above.

Marks (7). Both eigenvalues: 1. Basis for E_4: 1. Basis for E_1: 1. Valid basis justification: 1. Correct $\mathbf{P}$: 1. Matching $\mathbf{D}$: 1. Correct diagonalisation relation or verification: 1.

Common mistakes. Using the same eigenvector for both eigenvalues; placing eigenvalues in an order inconsistent with the columns of $\mathbf{P}$; writing $\mathbf{A} = \mathbf{P}^{-1}\mathbf{D}\mathbf{P}$; omitting the basis justification.

Section M2 — multivariable calculus

Question 8 — a vector chain rule

Part A — `MOCK-M2-Q08-PA` (M6) Method and conditions. Form the two Jacobians, evaluate the outer Jacobian at $\mathbf{G}(1,1,1)$, and multiply in chain-rule order.

$$\mathbf{J}_{\mathbf{F}}(u,v) = \begin{bmatrix} 2u & 1 \\ v & u \end{bmatrix} \quad (2 \times 2), \qquad \mathbf{J}_{\mathbf{G}}(x,y,z) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & z & y \end{bmatrix} \quad (2 \times 3).$$

Because $\mathbf{G}(1,1,1) = (2,1)^{\mathsf{T}}$,

$$\begin{aligned} \mathbf{J}_{\mathbf{F}\circ\mathbf{G}}(1,1,1) &= \mathbf{J}_{\mathbf{F}}(2,1)\mathbf{J}_{\mathbf{G}}(1,1,1) \\ &= \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 5 & 1 \\ 1 & 3 & 2 \end{bmatrix}. \end{aligned}$$

Final result.

$$\boxed{\mathbf{J}_{\mathbf{F}\circ\mathbf{G}}(1,1,1) = \begin{bmatrix} 4 & 5 & 1 \\ 1 & 3 & 2 \end{bmatrix}}.$$

Marks (4). Correct individual Jacobians and dimensions: 1. Correct evaluation point and evaluated outer Jacobian: 1. Dimensionally valid chain-rule product: 1. Correct product: 1.

Common mistakes. Evaluating $\mathbf{J}_{\mathbf{F}}$ at $(1,1)$ rather than $(2,1)$; reversing the multiplication order; using a 3×2 Jacobian for $\mathbf{G}$; making a row-by-column arithmetic error.

Question 9 — implicit differentiation

Part A — `MOCK-M2-Q09-PA` (M7) Method and conditions. To solve locally for y as a differentiable function of x, require $F_y \neq 0$. Then use $\mathrm{d}y/\mathrm{d}x = -F_x/F_y$.

$$F_y(x,y) = x + 2y.$$

At $(1,2)$, $F_y = 5 \neq 0$, so the local condition holds. Also,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{2x+y}{x+2y}, \qquad \left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{(1,2)} = -\frac{4}{5}.$$

For $\Delta x = 0.05$,

$$\Delta y \approx -\frac{4}{5}(0.05) = -0.04.$$

Final result. The required condition holds, the derivative is $-4/5$, and $\boxed{\Delta y \approx -0.04}$.

Marks (4). Correct local derivative condition and check: 1. Correct implicit-derivative formula: 1. Correct derivative value: 1. Correct first-order change: 1.

Common mistakes. Checking $F_x \neq 0$ when solving for y; omitting the minus sign; evaluating away from $(1, 2)$; treating -0.04 as the new level of y rather than its approximate change.

Question 10 — homogeneity and Euler's theorem

Part A — `MOCK-M2-Q10-PA` (M7) Method and conditions. Apply the definition with $t > 0$, then use Euler's theorem for a differentiable homogeneous function.

$$p(tx, ty) = (tx)^{2/5}(ty)^{3/5} = t^{2/5+3/5}p(x, y) = t\,p(x, y).$$

Thus p is homogeneous of degree 1. Euler's theorem gives

$$\boxed{xp_x(x, y) + yp_y(x, y) = p(x, y)}.$$

Taking $t = 2$ in the defining relation gives

$$p(2x, 2y) = 2p(x, y).$$

Final result. The function has degree 1, satisfies the displayed Euler relation, and doubling both inputs doubles p.

Marks (4). Correct definition-based scaling calculation: 1. Correct degree: 1. Correct Euler relation: 1. Correct doubling conclusion and interpretation: 1.

Common mistakes. Adding the inputs' exponents incorrectly; verifying only the special case $t = 2$ rather than the definition; omitting the input weights x and y from Euler's relation.

Section M3 — curvature and optimisation

Question 11 — unconstrained optimisation

Part A — `MOCK-M3-Q11-PA` (M8, M9) Method and conditions. Solve the stationary equations, use the Hessian for local classification, and use strict concavity on the whole convex domain for the global conclusion.

$$4 - 2x - y = 0, \qquad 6 - x - 2y = 0.$$

The unique stationary point is

$$(x^*, y^*) = \left(\frac{2}{3}, \frac{8}{3}\right).$$

The Hessian is

$$\mathbf{H}_f = \begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix}.$$

Its eigenvalues are -3 and -1, so it is negative definite. The point is a strict local maximum. Because f is strictly concave on the convex domain $\mathbb{R}^2$, it is also the unique global maximiser. Finally,

$$f\left(\frac{2}{3}, \frac{8}{3}\right) = \frac{28}{3}.$$

Final result.

$$\boxed{(x^*, y^*) = \left(\frac{2}{3}, \frac{8}{3}\right), \qquad f(x^*, y^*) = \frac{28}{3}}.$$

Marks (6). Correct stationary equations: 1. Correct stationary point: 1. Correct Hessian: 1. Valid negative-definite test and local classification: 1. Valid strict-concavity global argument: 1. Correct optimum value: 1.

Common mistakes. Treating the stationary equations as sufficient for a global maximum; classifying the Hessian by its diagonal entries alone; forgetting that the global conclusion needs concavity on the whole domain; mis-evaluating the cross term.

Question 12 — equality-constrained optimisation and the envelope theorem

Part A — `MOCK-M3-Q12-PA` (M10) Method and conditions. Use the stated constraint encoding, solve the first-order system on the positive domain, and apply the envelope theorem with that same encoding.

$$\mathcal{L}(x, y, \mu; a) = 2\log x + \log y + \mu(a - 2x - y).$$

The first-order conditions are

$$\frac{2}{x} - 2\mu = 0, \qquad \frac{1}{y} - \mu = 0, \qquad a - 2x - y = 0.$$

The first two equations imply $\mu = 1/x = 1/y$, so $x = y$. Feasibility then gives

$$x^*(a) = \frac{a}{3}, \qquad y^*(a) = \frac{a}{3}, \qquad \mu^*(a) = \frac{3}{a}.$$

For $a > 0$, the feasible set is convex and the objective is strictly concave on the positive orthant, so this is the unique global maximiser. The envelope theorem gives

$$V'(a) = \mathcal{L}_a(x^*, y^*, \mu^*; a) = \mu^*(a) = \frac{3}{a} > 0.$$

Under the encoding $a - 2x - y = 0$, the positive multiplier is the marginal increase in the optimised objective from one extra unit of a.

Final result.

$$\boxed{x^*(a) = y^*(a) = \frac{a}{3}, \qquad \mu^*(a) = V'(a) = \frac{3}{a}}.$$

Marks (6). Correct Lagrangian and first-order system: 1. Correct $x^*(a)$ and $y^*(a)$: 1. Correct multiplier: 1. Valid global sufficiency and uniqueness condition: 1. Correct envelope derivative: 1. Correct sign interpretation under the stated encoding: 1.

Common mistakes. Reversing the constraint encoding and retaining the same multiplier sign; losing the factor 2 in the x condition; asserting global uniqueness without concavity and convex feasibility; interpreting μ^* independently of how the constraint was written.

Question 13 — inequality constraints and a binding resource limit

Part A — `MOCK-M3-Q13-PA` (M11) Method and conditions. Express all inequalities as $g_j(x, y) \geq 0$, write the full course-convention Kuhn–Tucker system, solve the implied active set, and use strict concavity and convex feasibility for sufficiency.

Let

$$g_1(x, y) = x, \qquad g_2(x, y) = y, \qquad g_3(x, y) = 2 - x - y.$$

The Lagrangian is

$$\mathcal{L} = 3x + 2y - \frac{1}{2}x^2 - y^2 + \lambda_1 x + \lambda_2 y + \lambda_3(2 - x - y).$$

The complete Kuhn–Tucker conditions are

$$\begin{aligned}
&3 - x + \lambda_1 - \lambda_3 = 0,\\
&2 - 2y + \lambda_2 - \lambda_3 = 0,\\
&x \geq 0, \quad y \geq 0, \quad 2 - x - y \geq 0,\\
&\lambda_1, \lambda_2, \lambda_3 \geq 0,\\
&\lambda_1 x = 0, \quad \lambda_2 y = 0, \quad \lambda_3(2 - x - y) = 0.
\end{aligned}$$

Feasibility implies $x \leq 2$, and throughout the feasible set $\partial f/\partial x = 3 - x \geq 1 > 0$. If the resource constraint were slack, increasing x slightly would remain feasible and raise the objective. Therefore the resource constraint must bind, so $y = 2 - x$.

Temporarily suppose that both lower bounds are slack. Then $\lambda_1 = \lambda_2 = 0$, and stationarity gives

$$\lambda_3 = 3 - x = 2 - 2y.$$

Solving these equations yields

$$x^* = \frac{5}{3}, \qquad y^* = \frac{1}{3}, \qquad (\lambda_1^*, \lambda_2^*, \lambda_3^*) = \left(0, 0, \frac{4}{3}\right).$$

The resulting $x^* > 0$, $y^* > 0$ and $\lambda_3^* > 0$ verify the proposed active set rather than merely assuming it.

The resource constraint is active, while both non-negativity constraints are slack. The objective Hessian is

$$\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix},$$

which is negative definite. The feasible set is a convex compact triangle, so strict concavity makes the Kuhn–Tucker solution the unique global maximiser.

Final result.

$$\boxed{(x^*, y^*) = \left(\frac{5}{3}, \frac{1}{3}\right), \qquad (\lambda_1^*, \lambda_2^*, \lambda_3^*) = \left(0, 0, \frac{4}{3}\right).}$$

Marks (6). Correct Lagrangian: 0.5. Both stationarity equations: 0.5. Complete primal feasibility: 0.5. Dual feasibility and all complementary- slackness equations: 0.5. Correct active-set reasoning: 0.5. Correct (x^*, y^*): 1. Correct multipliers: 1. Correct active/slack report: 0.5. Valid global-uniqueness argument: 1.

Common mistakes. Writing the resource constraint with the opposite sign without changing the multiplier convention; omitting lower-bound complementarity; assuming the resource constraint binds without checking the resulting multipliers; reporting the right point with incomplete Kuhn–Tucker conditions.